

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered **real-time networking infrastructure** for device-to-cloud communication systems, focusing on distributed coordination and reliability under challenging network conditions.
- Developed **event-driven backend infrastructure** and APIs for platform systems, enabling seamless device <-> cloud interaction and data flow.
- Implemented **device identity, authentication, and security** protocols, ensuring secure communication and access control for a fleet of Linux-based controllers.
- Designed and deployed **observability, debugging, and production tooling** to enhance the monitoring and maintenance of real-world production systems.
- Managed **device update orchestration and rollout systems**, ensuring reliable and secure deployment of new firmware to a large-scale Linux device fleet.
- Contributed to guiding the **architecture of the platform** as it scales, making key decisions on infrastructure and system design for future growth.

SKILLS

Backend & Systems – Real-time networking, Distributed systems, Event-driven architecture, Device fleet management, API development, System reliability, Cloud infrastructure, Kubernetes, Networking, Linux systems.

Languages & Scripting – Python, Java, Spring Boot, SQL, Rust, TypeScript, Node.js, NestJS.

Databases & Storage – PostgreSQL, MySQL, Oracle, S3.

Cloud & Containerization – AWS (EC2, Lambda, CloudWatch), Docker.

DevOps & Tools – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace.

PROJECTS

Secure Device Fleet Management Platform (HCL Technologies)

- Developed a **Python-based backend service** on AWS Lambda to manage device registration and authentication for a fleet of 5,000+ Linux controllers, integrating short-lived tokens issued by a NestJS auth service.
- Implemented **mutual-TLS device identity and certificate rotation** mechanisms, drastically reducing credential drift incidents across rolling fleet updates and enhancing overall system security.
- Engineered **real-time communication channels** using MQTT for device telemetry and command-and-control, ensuring reliable data exchange even on unreliable networks.

Distributed System Reliability & Observability Tools (HCL Technologies)

- Built a **distributed coordination service** using PostgreSQL and Python to manage state and prevent race conditions across concurrent device operations, improving system stability by 18%.
- Created **automated diagnostic scripts** in Python and shell for Linux systems, enabling faster root cause analysis of P1/P2 incidents and reducing average resolution time by 25%.
- Designed and implemented **event-driven data pipelines** using Kafka to ingest logs and metrics, feeding into a centralized observability platform (Splunk) for enhanced debugging.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)